

Machine Learning Assignment 01

(100 Marks)

Objective

The goal of this project is to analyze a car dataset, preprocess the data, apply **KNN Regression**, **Multilinear Regression** and **Decision Tree Regression** and evaluate their performance in predicting car prices.

Mandatory (Must Provide Insights with each question of section.)

Section 1: Data Exploration (EDA) – 25 Marks

1.1 Loading the Data (5 Marks)

- Load the dataset using Pandas.
- Display the first five rows.
- Check for missing values in each column.
- Identify numerical and categorical features, and check their data types.

1.2 Statistical Summary and Distributions (8 Marks)

- Generate a summary report using `.describe()` for numerical columns.
- Plot histograms for numerical features.
- Identify outliers using boxplots for at least three numerical columns.

1.3 Relationships Between Features (12 Marks)

- Compute and visualize the correlation matrix. Identify features highly correlated with **price**.
- Use a pivot table to find the average price for different fuel types.
- Analyze the average price for different car brands using `groupby()`.
- Examine how body style impacts car price with suitable plots.
- Determine the most common engine type and represent it using a bar chart.

Section 2: Data Preprocessing – 25 Marks

2.1 Handling Missing Values (8 Marks)

- Identify columns with missing values and calculate the percentage of missing data.
- Decide on a strategy for missing numerical values (mean/median/mode).
- Decide on a strategy for missing categorical values.
- Apply missing value handling and verify no NaNs remain.

2.2 Encoding Categorical Features (6 Marks)

- List all categorical features that require encoding.
- Apply **One-Hot Encoding** or **Label Encoding** (justify your choice).
- Verify encoding by checking the transformed dataset.

2.3 Feature Scaling and Selection (6 Marks)

- Identify features that need scaling (especially for KNN). Explain why.
- Apply **standardization** or **normalization** to necessary features.
- Select relevant features for modeling based on correlation with price.

2.4 Splitting Data (5 Marks)

- Split the dataset into training (80%) and testing (20%) sets.
- Explain why keeping a separate test set is important.

Section 3: Model Building – 30 Marks

3.1 Applying Regression Models

- **Multilinear Regression (8 Marks):** Train a linear regression model and record **MSE** and **R² Score**.
- **Decision Tree Regression (8 Marks):** Train a Decision Tree model. Compare its performance with Multilinear Regression.
- **KNN Classifier (8 Marks):** Train a KNN model. Experiment with different values of k . Compare performance with other models.

3.2 Evaluating Model Performance (6 Marks)

- Create a table comparing **MSE** and **R² Score** for all three models.
- Identify which model performed best and explain why.
- If any model overfits, explain how it can be improved (hyperparameter tuning, pruning, regularization, scaling).

Section 4: Reporting & Insights – 20 Marks

4.1 Summary of Findings (20 Marks)

- Key insights gained from EDA about car prices.
- Features with the most impact on price prediction.
- Challenges faced during preprocessing and modeling.
- Additional steps you would take if you had a larger dataset with more features.

Notes

- KNN Regression requires **feature scaling**.
- Decision Tree Regression can handle categorical features but may **overfit**; consider max depth tuning.
- Multilinear Regression assumes **linear relationships** and may not capture complex patterns.